

RÉPUBLIQUE DU CAMEROUN
Paix-Travail-Patrie

MINISTÈRE DE L'ENSEIGNEMENT SUPÉRIEUR

COMMISSION NATIONALE D'ORGANISATION DE L'EXAMEN
HIGHER NATIONAL DIPLOMA EXAM (HND)

REPUBLIC OF CAMEROON
Peace-Work-Fatherland

MINISTRY OF HIGHER EDUCATION

NATIONAL COMMISSION FOR THE ORGANISATION OF HIGHER
NATIONAL DIPLOMA EXAM (HND) EXAM

National Exam of Higher National Diploma-New program – 2021 Session

Spécialty/option :Software Engineering (SWE)

Paper :Information System

Duration : 3Hours

Credits : 6

Instruction: Answer all questions

SECTION A: SYSTEM ARCHITECTURE /40 marks

1. What is network architecture? (2 marks)
2. List and describe the two types of widely used network architectures illustrating the diagram of each (1+4x2=9 marks)
3. Give the meaning of OSI as designed by the International standard organization. List and explain the different layers of the OSI model in descending order. And precise the unit data at each layer (1+7+3.5=11 marks)
4. In computing most of time people are using the word cloud. Explain what is a cloud? (2 marks)
5. What is Cloud Computing? Give an illustration of it (2+2=4 marks)
6. From the experience that you have acquired when learning cloud computing, list and explain 4 benefits of Cloud Computing? (1.5x4=6 marks)
7. Answer true or false on the following questions (0.5x12=6 marks)
 - a. Hardware as a Service is not a major type of cloud computing usage
 - b. Virtual Machine Ware (VMware) is an example of software Service
 - c. API stand for Analysis Programming Interface
 - d. A Cloud application relies on Remote Servers
 - e. A Cloud app is Hardware program
 - f. A Google Apps Engine is a type of CaaS
 - g. A good cloud computing network cannot be adjusted to provide bandwidth on demand
 - h. IBM Bluemix is a cloud platform as a service developed by IBM
 - i. Cloud Service has a many to many Relationship with their customers
 - j. Google doc is a type of Cloud computing
 - k. Cloud providers provide cloud services to cloud users.
 - l. Cloud applications can function offline

SECTION B: OPERATING SYSTEMS. (40 marks)

1. What is an operating system (5marks)
2. What is memory management? (5marks)
3. State the activities performed by the operating system for processor management (5marks)
4. Differentiate between general purpose operating system and mobile operating system (5marks)
5. List the five (5) components of information system (10marks)
6. Why study the human use of computers systems? (5marks)
7. List the four (4) key components of android architecture (5marks)

SECTION C: PROJECT MANAGEMENT AND LEGAL REGULATIONS (20 marks)

Part 1: Short Structurals(10marks)

1. Define the following terms as used in project management? (2marks)
 - a. Project portfolio
 - b. Project manager
 - c. Project plan
 - d. Project charter
2. List and explain at least 2 project management knowledge areas (2marks)
3. List and explain 2 major phases of a project management life cycle (2marks)
4. Who is an intervenor in legal terms? (1mark)
5. List and explain 3 types of intervenors you know (3marks)

Part 2: Essay (10 mks)

1. You are the new project manager for the construction of 2 classrooms at ABC Higher Institute. Draft a project plan showing all the phases, task activities, resources, time allocation, resource allocation, and necessary charts for the project. From your time frames, calculate the optimistic, pessimistic and most likely estimates for the first and last activity on your plan (5marks)
2. Using your knowledge of market research, suggest to your school a marketing research plan that allows them increase the number of international student admission (5 marks)